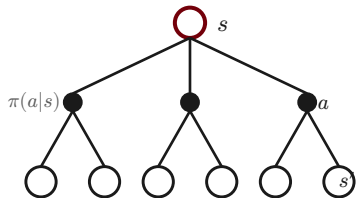
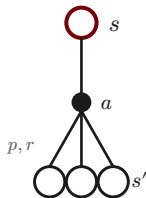


v_π backup



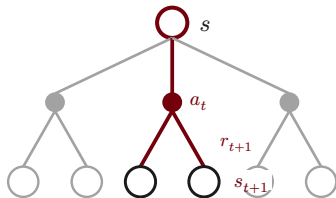
average over **both** actions (π)
and outcomes (p)

q_π backup



action a **fixed**; average
over outcomes (p) only

sample (TD) backup



follow **one** sampled
transition — no averaging

$$v_\pi(s) = \sum_a \pi(a|s) \sum_{s',r} p(s',r | s,a) [r + \gamma v_\pi(s')]$$

the backup diagram is this expectation drawn as a tree: outer sum = action fan, inner sum = outcome fan, leaves = $v_{\pi(s')}$.

○ state node (open circle)

● action node (filled circle)

○ node being updated

↑ backup direction